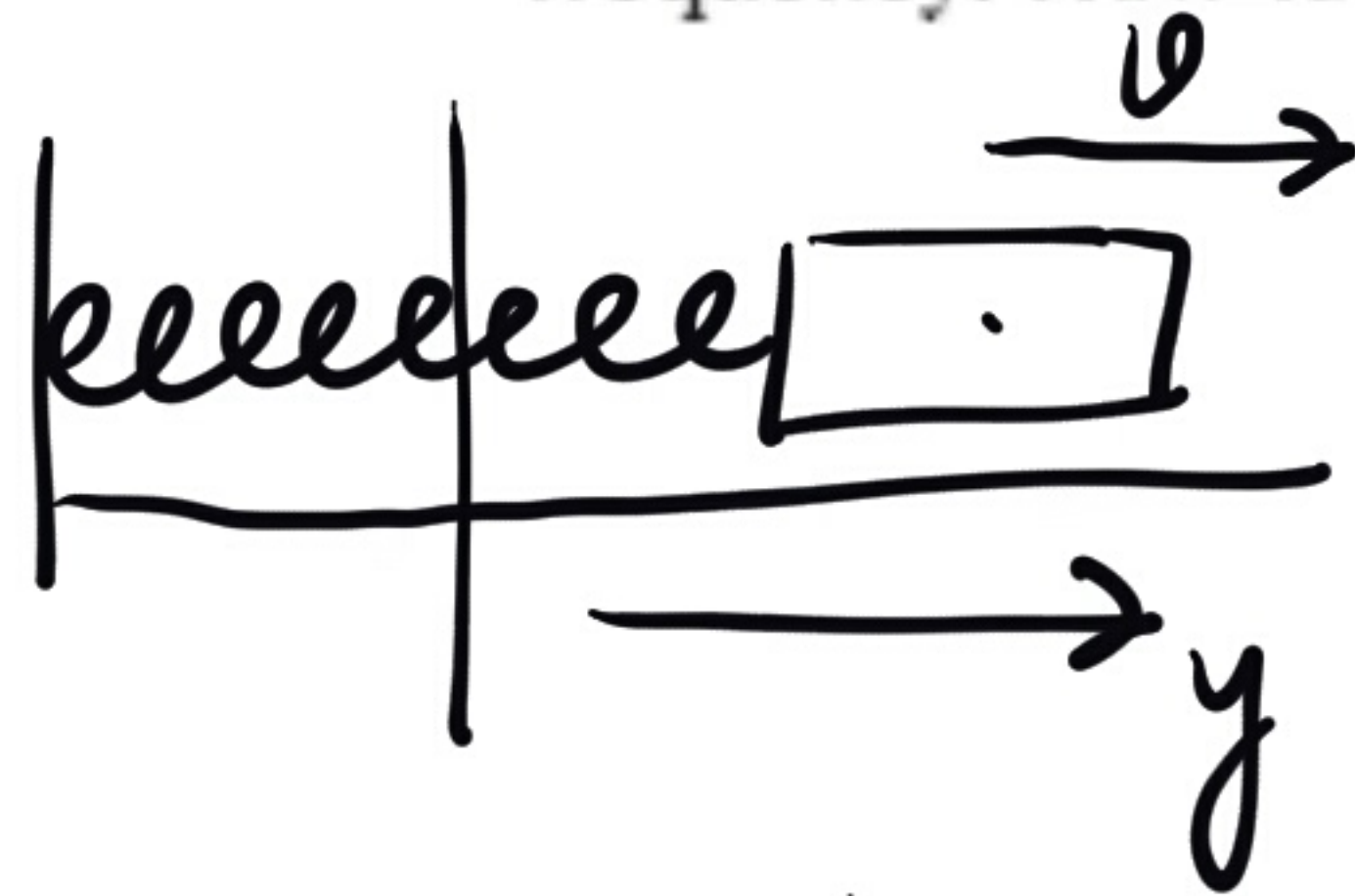


Example 1

A $1/8$ -kg mass is attached to a spring with stiffness $k = 16$ N/m, as depicted in Figure 4.1. The mass is displaced $1/2$ m to the right of the equilibrium point and given an outward velocity (to the right) of $\sqrt{2}$ m/sec. Neglecting any damping or external forces that may be present, determine the equation of motion of the mass along with its amplitude, period, and natural frequency. How long after release does the mass pass through the equilibrium position?

(p 215)



$$m = \frac{1}{8}, \quad k = 16, \quad y(0) = \frac{1}{2}, \quad y'(0) = \sqrt{2}$$

$$b = 0$$

$$\frac{1}{8} y'' + 16y = 0 \Rightarrow y'' + 128y = 0$$

$$r^2 + 128 = 0 \Rightarrow r_{1,2} = \pm 8\sqrt{2}i$$

"ω"

$$y(t) = C_1 \sin 8\sqrt{2}t + C_2 \cos 8\sqrt{2}t$$

$$y'(t) = C_1 8\sqrt{2} \cos 8\sqrt{2}t - C_2 8\sqrt{2} \sin 8\sqrt{2}t$$

$$y(0) = \frac{1}{2} \Rightarrow \frac{1}{2} = C_2, \quad y'(0) = \sqrt{2} \Rightarrow \sqrt{2} = C_1 8\sqrt{2} \Rightarrow C_1 = \frac{1}{8}$$

$$\Rightarrow y = \frac{1}{8} \sin 8\sqrt{2}t + \frac{1}{2} \cos 8\sqrt{2}t = \frac{\sqrt{17}}{8} \sin(8\sqrt{2}t + \varphi)$$

$$A = \sqrt{\frac{1}{64} + \frac{1}{4}} = \frac{\sqrt{17}}{8}$$

$$\varphi = \arctan 4$$

(I quad)

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{8\sqrt{2}} = \frac{\pi}{4\sqrt{2}}, \quad \gamma = \frac{1}{T} = 4\sqrt{2}/\pi$$

$y(t) = ?$
 $A, T, \gamma,$

$t_0 = ?$

$y(t_0) = 0$

$$y(t_0) = \frac{\sqrt{17}}{8} \sin(8\sqrt{2}t_0 + \varphi) = 0$$

$$\Rightarrow \sin(8\sqrt{2}t_0 + \varphi) = 0$$

$$8\sqrt{2}t_0 + \varphi = \pi$$

$$8\sqrt{2}t_0 = \pi - \varphi$$

$$t_0 = \frac{\pi - \varphi}{8\sqrt{2}}$$

